STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 12: Kinetic Theory

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. What is an ideal gas model?
2. Ideal gas equation?
3. Mean translational kinetic energy?
4. Origin of gas pressure?
5. Monatomic degrees of freedom?
6. Effect of lowering pressure on mean free path?
7. Molar C_v of monatomic gas?
8. Rms speed relation?

Section B 2-mark questions - 6 x 2 = 12 marks

9. At fixed T, volume halves: pressure?
10. 27 deg C in Kelvin?
11. Mean kinetic energy at 300K, k=1.38 x 10-^2^3?
12. Temperature becomes 4T: rms speed factor?
13. Internal energy of n mol monatomic gas?
14. Two main ideal gas assumptions?

Section C 3-mark questions - 4 x 3 = 12 marks

15. Rms speed of M=.028,T=300K,R=8.31?
16. mol monatomic gas warms 20K, C_v=3R/2: DeltaU?
17. At fixed T, pressure triples: volume factor?
18. State equipartition prediction for monatomic gas.

Section D 4-mark questions - 2 x 4 = 8 marks

19. Compare rms speeds for molar masses .004 and .016 kg/mol at same T.
20. f=5, n=2 mol, T=300K: internal energy?